USED PAPERS AUG/SEP-2021

Code No: 156EA **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August/September - 2021
RENEWABLE ENERGY SOURCES
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) How are electricity generation costs calculated for existing plants and new plants?
b) Elaborate the factors to be considered in the selection of site for wind energy. [8+7]

2.a) Write a short note on management options on demand side and supply side.
b) What is wind power? Derive an equation for wind power. [7+8]

3.a) Explain how electricity is generated by photovoltaic effect.
b) Explain the constructional features of proton exchange membrane fuel cells. [7+8]

4.a) How are solar panels connected in series and in parallel.
b) What are the advantages and disadvantages of fuel cells? [7+8]

5. Give the mathematical description of the self-excitation process in an induction generator. [15]

6. Explain the speed and voltage control of induction generators. [15]

7.a) What are advantages and disadvantages of lead acid batteries?
b) Explain compressed air energy storage system with the help of a neat sketch. [7+8]

8.a) Explain the concept of instantaneous active and reactive power control.
b) What are the different codes and standards for Interconnection of alternate energy sources with the grid? [7+8]

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